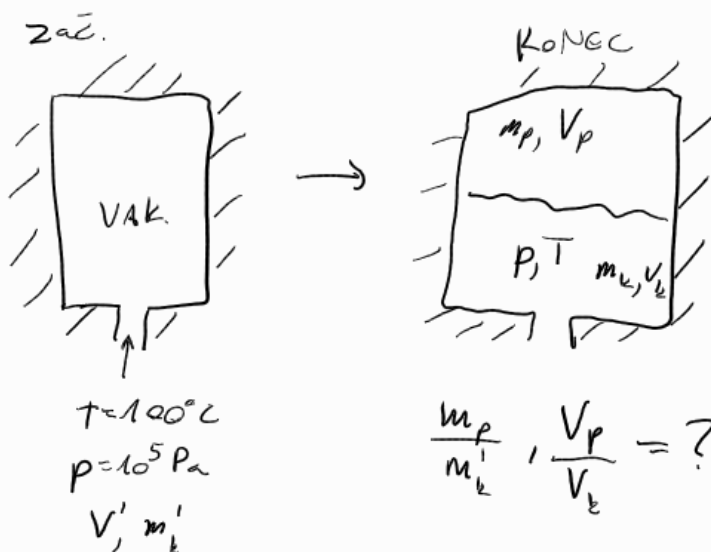


2.8) Vdorude v izolirano evakuirano posodo



$$\frac{1}{\rho_p} = 1,673 \frac{\text{m}^3}{\text{kg}}$$

$$\frac{1}{\rho_k} = 1,043 \frac{\text{dm}^3}{\text{kg}}$$

$$\Delta U = Q + W$$

$$W = pV$$

$$V_p - V_k \approx V_{\text{pure}}$$

$$\Delta U = m_p q_i - p \Delta U = m_p q_i - \frac{m_p}{M} R T = m_p \left(q_i - \frac{RT}{M} \right) = pV = p \frac{m'_k}{\rho_k}$$

$$\frac{m_p}{m'_k} = 5 \cdot 10^{-5}$$

$$\frac{V_p}{V_k} = \frac{\rho_k m_p}{\rho_p m'_k} = 9,08$$

$m_k \approx m'_k$

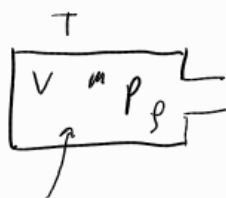
2.5) Iz jeklenke uide $\frac{1}{2}$ vsebine

$$V = 5 \text{ dm}^3$$

$$p = 10^6 \text{ Pa}$$

$$T = 20^\circ\text{C} = \text{konst}$$

$$Q = ?$$



$$dm = -d\tilde{m}$$

$$dU = dQ + dW = 0$$

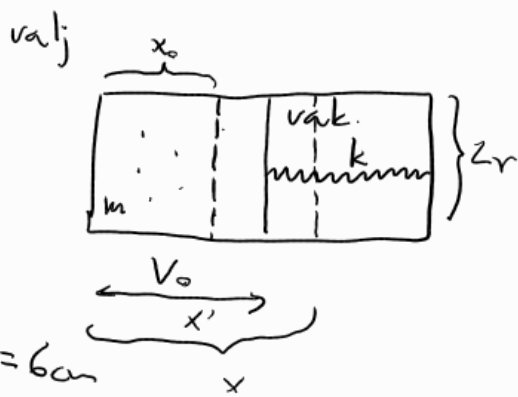
$T = \text{konst.}$

$$dQ = -dW = p dV$$

$$Q = \int p dV = \int_m^{m/2} p \frac{d\tilde{m}}{\rho} = \int_m^{m/2} \frac{\rho R T d\tilde{m}}{\rho M} =$$

$$= \frac{RT}{M} \int_{m/2}^m d\tilde{m} = \frac{mRT}{2M} = \frac{pV}{2} = 2,5 \text{ kJ}$$

$$\rho = \frac{pM}{RT}$$



$$z_r = 6 \text{ cm}$$

$$m = 1 \text{ g}$$

$$k = 10^3 \text{ N/m}$$

$$V_0 = 1 \text{ dm}^3 \rightarrow x_0 = \frac{V_0}{A}$$

$$T' = 20^\circ \text{C} \rightarrow T = 50^\circ \text{C}$$

$$M = 29 \frac{\text{kg}}{\text{kmol}}$$

$$c_v = 718 \frac{\text{J}}{\text{kg K}}$$

$$Q = ?$$

$$F_p = F_v$$

$$pA = k(x - x_0)$$

$$p = \frac{mRT}{MV} = \frac{m}{M} \frac{RT}{xA}$$

$$\rightarrow \frac{mRT}{MA} = kx^2 - kxx_0$$

$$x_{1,2} = \frac{k_0 x_0 \pm \sqrt{k_0^2 x_0^2 + 4k \frac{mRT}{M}}}{2k}$$

$$= 95286 \text{ m}$$

$$\Delta U = Q + W = Q - \Delta W_{pr}$$

$$Q = mc_v(T - T') - \frac{k}{2} ((x - x_0)^2 - (x' - x_0)^2) = 23,64 \text{ J}$$

Delo v magn. sist.

$$\text{TD spr.: } H, T \rightarrow dW_m = \mu_0 H d(VB)$$

$$H, M, T \rightarrow dW_m = \mu_0 H d(VM)$$

$$\text{primer: paramagnet } M = C \frac{H}{T} \text{ (Curiejev zakon)}$$

delo pri magnetenju ($T = \text{konst.}$)

$$W_m = \mu_0 V \int_{H_1}^{H_2} H dM = \mu_0 V \frac{C}{T} \int_{H_1}^{H_2} H dH$$

$$(H = \text{konst.})$$

$$W_m = \dots = \mu_0 V H^2 C \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$